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Conjugate addition of arylsilanes to unsaturated carbonyl compounds catalyzed by rhodium in air and water

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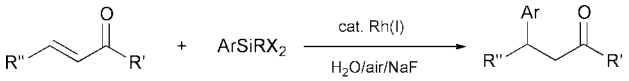
Received (in Cambridge, UK) 10th July 2001, Accepted 19th September 2001
First published as an Advance Article on the web 22nd October 2001

$ArSiCl_3$ and Ar_2SiCl_2, believed to be unstable in aqueous media, reacted efficiently with α,β-unsaturated ketones and esters in air and water (in the presence of sodium fluoride and a rhodium catalyst), giving good to excellent yields of the desired conjugate addition products.

Currently, there is a wide interest in searching for alternative media and processes for chemical and organic synthesis. As a result of the natural abundance of water as well as the inherent advantages of using water as a solvent, interest has been growing in studying organic reactions in water.[1] Recently, in our ongoing 'green' chemistry research program, namely, the use of environmentally friendly reagents, solvents, and processes for organic reactions, we have been interested in developing transition-metal catalyzed carbon–carbon bond formations under the quasi-natural conditions of air and water.[2] Such reactions have the advantages of simplifying protection–deprotection steps, easy recycling of the catalyst solution, and convenient conducting of the reaction. A general requirement for performing such reactions is that the reagent is air- and water-stable, readily available, and less toxic (nontoxic if possible).

Conjugate addition of organometallic reagents to α,β-unsaturated carbonyl compounds is an important means for C–C bond formation in organic synthesis.[3] This reaction was once limited to organocopper chemistry but now is possible with other organometallic compounds through the use of transition metal catalysts.[4] Recently we found that, in the presence of a rhodium catalyst, α,β-unsaturated esters and ketones reacted with arylbismuth, aryltin (and vinyltin), and aryllead reagents in aqueous media to give the corresponding conjugate addition products in an air atmosphere.[5] The method has been used in synthesizing natural and unnatural α-amino acids.[6] However, organotin, organolead, and organobismuth reagents are rather toxic in spite of the fact that they are very efficient in the quasi-natural conditions of water and air. Therefore we felt it was highly desirable to search for the less toxic reagents instead of tin, lead, or bismuth.

Silicon has drawn the attention of chemists for many years due to its low cost, large abundance, and nontoxic properties. The application of silicon and organic silicon compounds for synthetic purposes has been growing in scope due to their relatively high stability and the ability to tailor chemo-, regio-, and stereoselectivities by combining these reagents with appropriate catalysts. Widely recognized reactions attributed to the use of silicon include organosilane based reduction chemistry,[7] cross-coupling reactions,[8] allylsilanes chemistry,[9] as protecting groups,[10] and the chemistry of silyl enolates.[11] Recently, transition-metal catalyzed conjugate additions involving allylsilanes were studied under an inert gas atmosphere and anhydrous conditions.[12] However, conjugate addition with arylsilicon compounds has not been reported due to their lower reactivity, especially in the presence of air and water. Herein we wish to report a highly efficient conjugate addition reaction with arylsilanes in air and water in the presence of sodium fluoride and catalyzed by rhodium catalysts (eqn. 1).



(1)

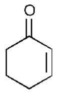
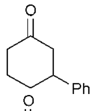
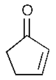
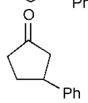
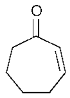
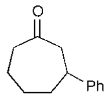
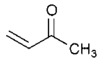
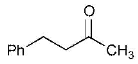
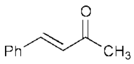
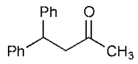
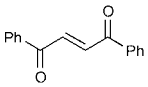
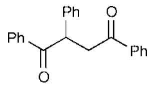
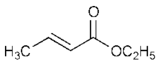
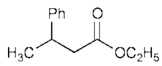
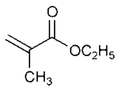
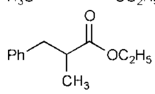
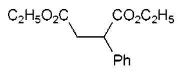
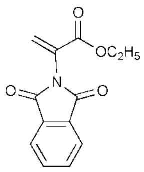
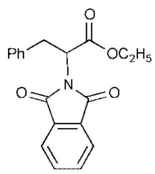
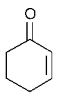
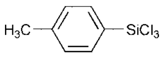
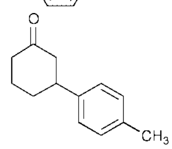
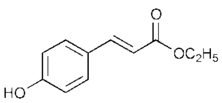
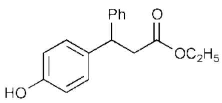
Initially, phenyltrimethylsilane was chosen to react with 2-cyclohexen-1-one in the presence of a catalytic amount of $(COD)_2RhBF_4$ in water because of its high stability in aqueous media. However no reaction was observed, even after refluxing for several days in the presence of sodium fluoride. Knowing that arylsilyl halides undergo transmetalation with palladium complexes with the aid of fluoride ions *via* five-coordinate silicates in the cross-coupling reaction with aryl- or alkenylhalides,[13] we selected phenyltrichlorosilane as a phenylating reagent despite the fact that phenyltrichlorosilane was thought to be unstable in water. Subsequently, it was discovered that, when cyclohex-2-en-1-one was stirred with phenyltrichlorosilane and a catalytic amount of $(COD)_2RhBF_4$ in the presence of an excess amount of NaF at refluxing conditions overnight in water, a smooth reaction occurred to give the conjugate addition product in 54% isolated yield. Unlike the triphenylbismuth reaction, virtually no biphenyl product was observed with the silicon reagent. No reaction was observed without the catalyst or fluoride. At room temperature the reaction also ceased to proceed completely. Interestingly, the use of different fluoride salts also affected the reaction progress. Fluoride salts such as lithium fluoride, sodium fluoride and potassium fluoride were tested, and sodium fluoride appeared to be the most effective for this reaction. Among the rhodium catalysts screened, $(COD)_2RhBF_4$ and $(COD)_2RhCl$ were effective while others such as $[(CO)_2RhCl]_2$, $(PPh_3)_3RhCl$, and $(C_2H_4)_2Rh(acac)$ were not. It is noteworthy that the yield of this reaction was dramatically improved to almost 100% when (4 equivalents) Ph_2SiCl_2 was used instead of $PhSiCl_3$ (Table 1). The use of 1 equivalent of the silicon reagent decreased the yield somewhat (67%). It is noteworthy that essentially identical results were obtained with either Ph_2SiCl_2 or $Ph_2Si(OH)_2$ as the phenylating reagent, which implies that hydrolysis of the silyl chloride does

Table 1 Effects of reagents and catalyst on conjugated addition of cyclohex-2-en-1-one

$ArMR_n$	Conditions	Yield (%)
$PhSiCl_3$	H_2O/reflux/$(COD)_2RhBF_4$/NaF	54
Ph_2SiCl_2	H_2O/reflux/$(COD)_2RhBF_4$/NaF	97
$PhMeSiCl_2$	H_2O/reflux/$(COD)_2RhBF_4$/NaF	95
$PhSiMe_3$	H_2O/reflux/$(COD)_2RhBF_4$/NaF	0
Ph_2GeCl_2	H_2O/reflux/$(COD)_2RhBF_4$/NaF	0
$PhGeCl_3$	H_2O/reflux/$(COD)_2RhBF_4$/NaF	0
$PhSiCl_3$	H_2O/reflux/$(COD)_2RhBF_4$/KF	48
$PhSiCl_3$	H_2O/reflux/$(COD)_2RhBF_4$/LiF	Trace
$PhSiCl_3$	H_2O/reflux/$(COD)_2RhBF_4$/NaOH	< 10
$PhSiCl_3$	H_2O/reflux/$(COD)_2RhBF_4$	0
Ph_2SiCl_2	H_2O/rt/$(COD)_2RhBF_4$/NaF	Trace
Ph_2SiCl_2	H_2O/reflux/$[(CO)_2RhCl]_2$/NaF	0
Ph_2SiCl_2	H_2O/reflux/$(CH_2{=}CH_2)_2$ Rhacac/NaF	0
Ph_2SiCl_2	H_2O/reflux/$(PPh_3)_3RhCl$/NaF	0
Ph_2SiCl_2	H_2O/reflux/NaF	0

 DOI: 10.1039/b106768k

Table 2 Conjugated addition with arylsilanes in air and water[a]

Entry	Carbonyl derivative	Silane	Product	Yield (%)
1		Ph_2SiCl_2		97
2		Ph_2SiCl_2		81
3		Ph_2SiCl_2		95
4		Ph_2SiCl_2		34
5		Ph_2SiCl_2		87
6		Ph_2SiCl_2		96
7		Ph_2SiCl_2		71
8		Ph_2SiCl_2		61
9		Ph_2SiCl_2		85
10		Ph_2SiCl_2		94
11				47
12		Ph_2SiCl_2		61

[a] All reactions were carried out at refluxing conditions with isolated yields after column chromatography on silica gel.

not affect the reaction. Under the same reaction conditions, no reaction was observed with analogous germanium compounds. With the optimal reaction conditions in hand, several α,β-unsaturated carbonyl compounds were examined with Ph_2SiCl_2 (or $TolSiCl_3$ for entry 11) under these conditions (Table 2). Both ketones (linear and cyclic) and esters were effective as the electron-withdrawing functional groups. Compounds bearing a hydroxy group reacted as expected without the need of protection.

The following procedure is representative. A mixture of cyclohex-2-en-1-one (0.25 mmol), diphenyldichlorosilane (253 mg, 1.0 mmol), NaF (210 mg, 5.0 mmol) and $(COD)_2RhBF_4$ (5.6 mg, 0.013 mmol) in 5 mL of water was stirred at 100 °C for 12 h under an air atmosphere. The reaction mixture was extracted with ethyl acetate. After the combined solvent was dried and concentrated *in vacuo*, the residue was purified by silica gel chromatography eluting with hexanes–ethyl acetate (5 : 1) to give 42.2 mg (97%) of the desired product.

In conclusion, a highly effective conjugate addition of diphenyldichlorosilane to α,β-unsaturated ketones and esters was achieved by using a rhodium catalyst in the presence of sodium fluoride in air and water. The scope (such as asymmetric synthesis), mechanism, and synthetic applications of this reaction are currently under investigation.

We are grateful to the NSF (CAREER) and the NSF-EPA Joint Program for a Sustainable Environment for partial support of this research.

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